Longitudinal Analysis of Psychiatric Symptoms in Alzheimer's Disease

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Study description

- **Population:** 1,253 patients with Alzheimer's disease enrolled in **25 clinical trials**.
- **Outcome:** Brief Psychiatric Rating Scale (BPRS), measured annually from baseline to year 6
  - Up to **7 measurements per patient** (baseline + 6 follow-up years)
  - Higher BPRS = more severe psychiatric symptoms
- **Design:** Unbalanced longitudinal data
  - Not all patients are observed at all time points
  - Total number of BPRS observations: **6,220**
- **Baseline covariates:**
  - Demographic: sex, age, education
  - Clinical: BMI, ADL score, living situation (home vs nursing home), job status (paid vs no paid job)
  - Disease severity / biomarkers: baseline CDR-SB, amyloid PET (ABPET0), tau PET (TAUPET0)

Missing data analysis

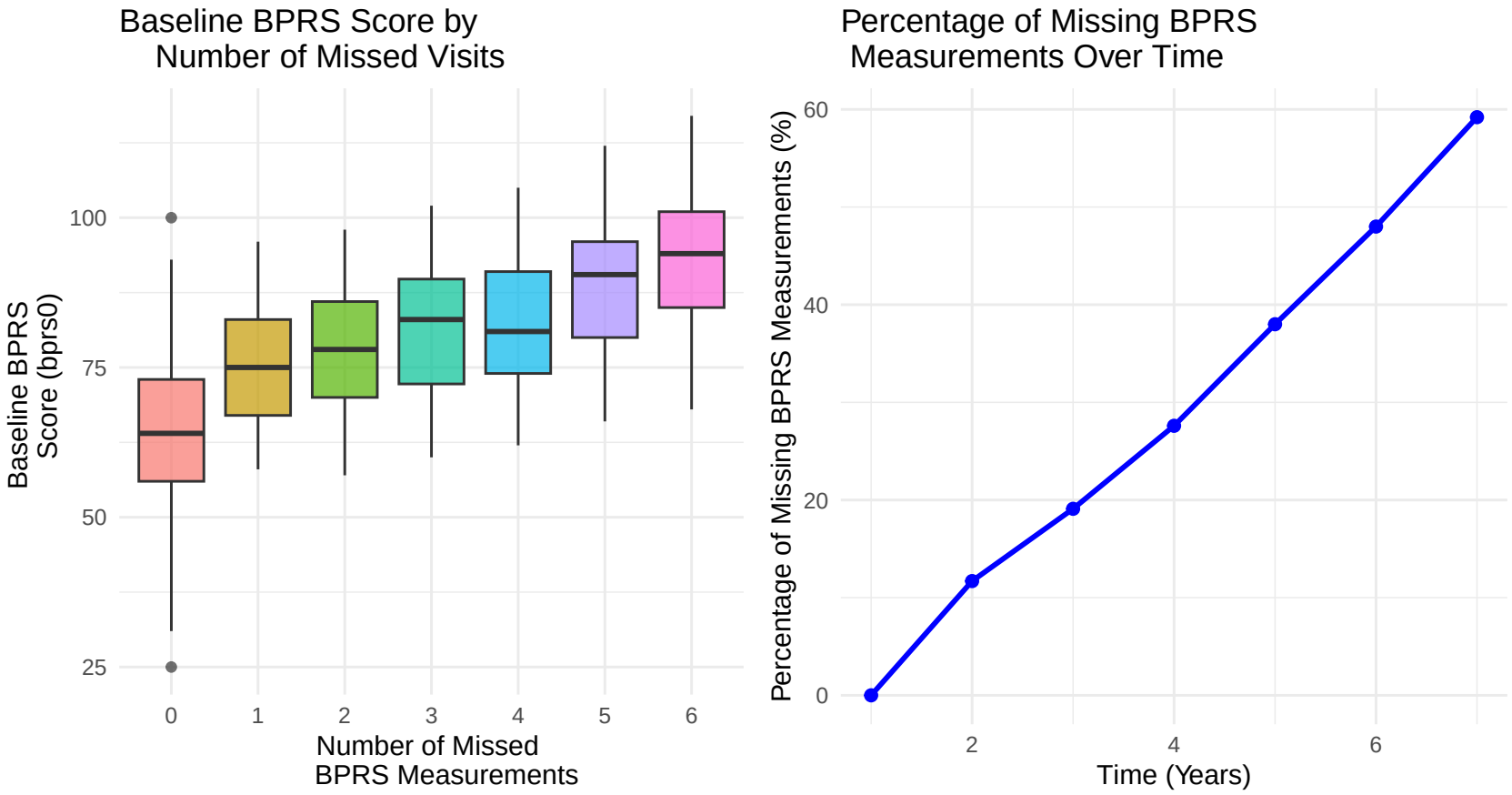

Figure 1: Missingness Analysis: Baseline BPRS by Number of Missed Visits (left) and Percentage Missing Over Time (right)

BPRS over time

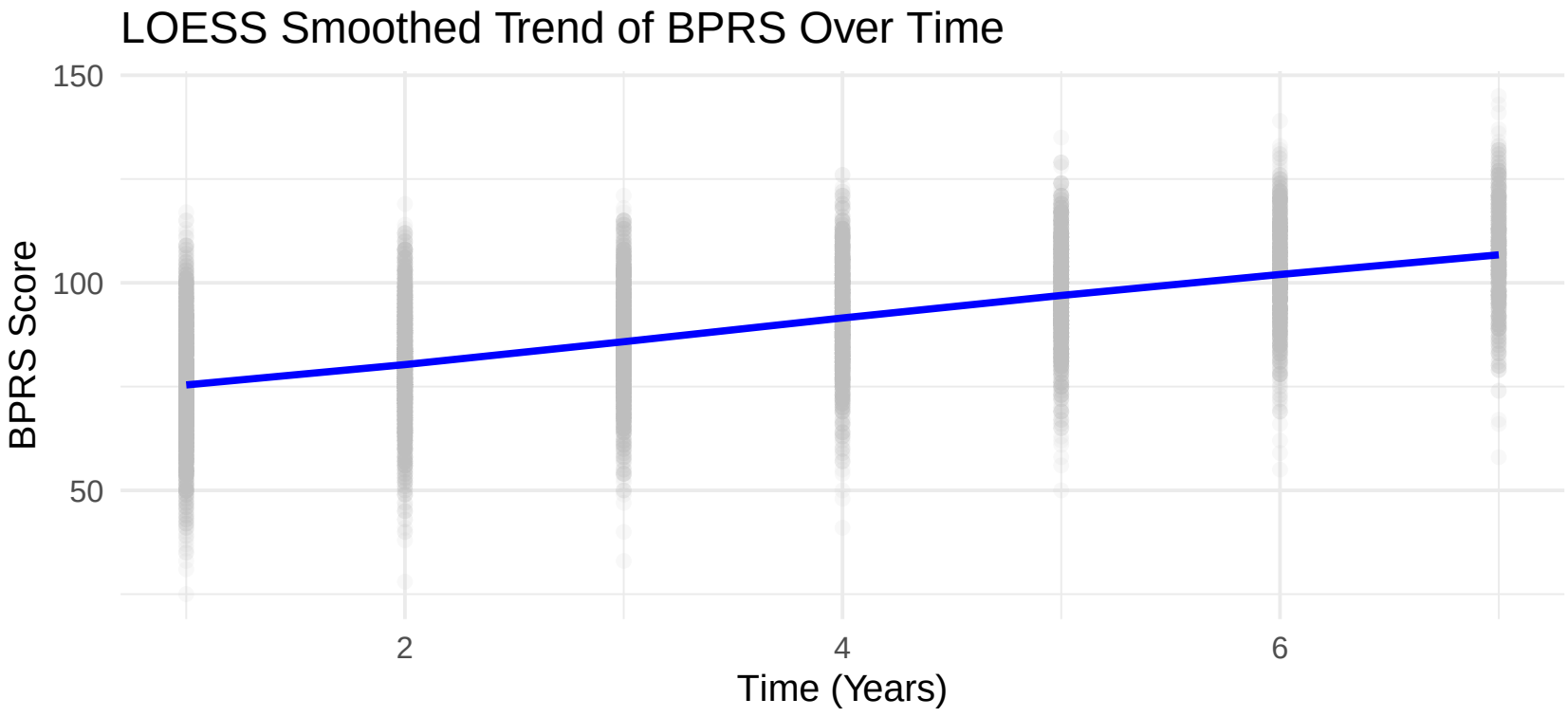

Figure 2: Average BPRS Scores Over Time

Individual profiles

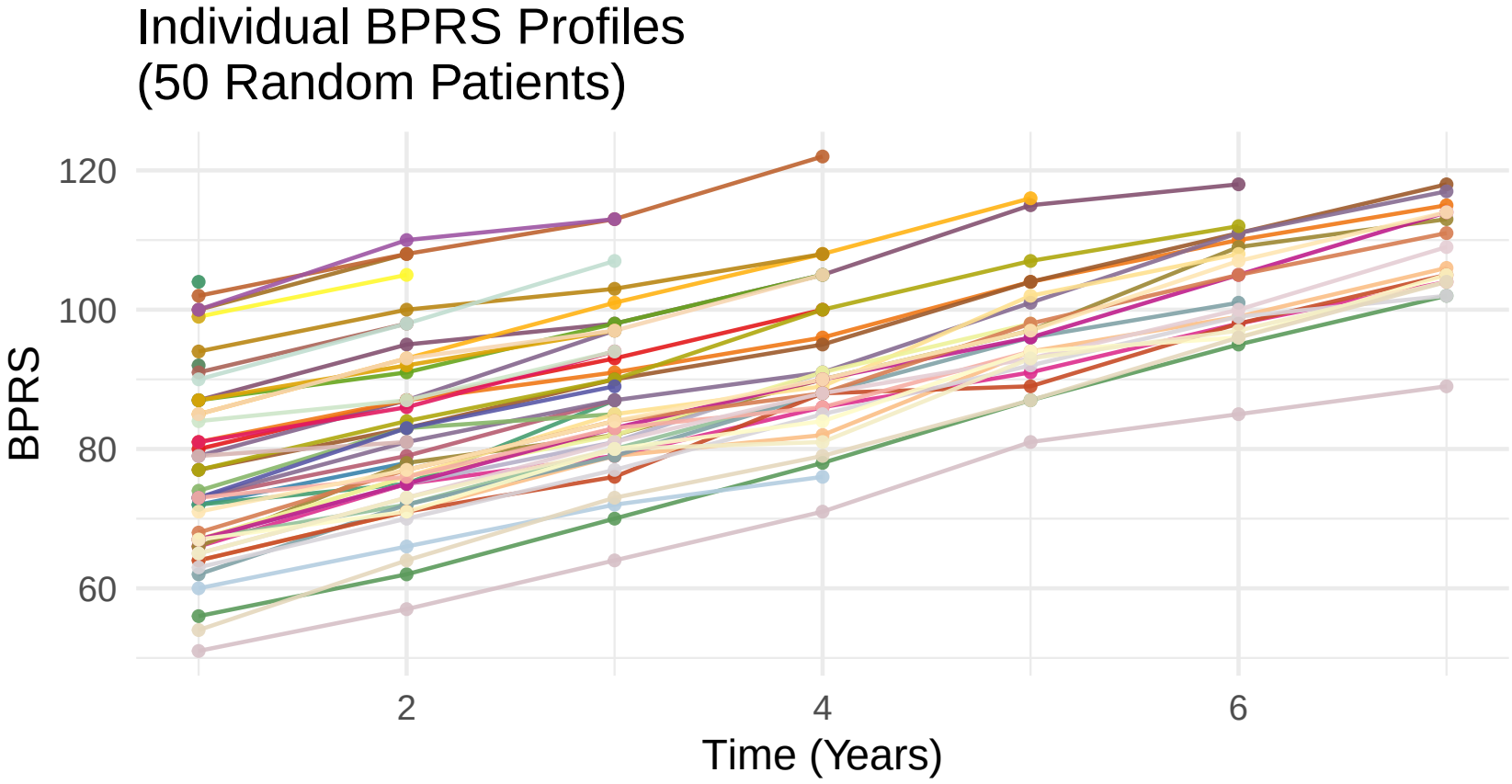

Figure 3: Spaghetti Plots of BPRS Over Time by Clinical Center

- These 50 individual profiles show that patients start at very different BPRS levels but almost all worsen over time, with broadly similar upward slopes.

Separate Linear Models for Each Follow-up Year

Table 1: Separate Linear Models for Each Follow-up Year

covariate	Year0	Year1	Year2	Year3	Year4	Year5	Year6
Intercept	-122.72 [-126.28, -119.16]	-119.78 [-126.03, -113.52]	-105.57 [-120.15, -90.99]	-104.92 [-111.33, -98.51]	-101.46 [-109.63, -93.29]	-95.55 [-106.07, -85.02]	-87.73 [-101.67, -73.79]
Baseline Amyloid PET	0.34 [-0.07, 0.74]	0.20 [-0.29, 0.69]	0.03 [-0.61, 0.68]	-0.13 [-0.98, 0.71]	-0.03 [-1.18, 1.13]	-0.10 [-1.74, 1.54]	-1.11 [-3.75, 1.54]
Activities of Daily Living Score	2.01 [1.92, 2.11]	2.17 [2.05, 2.29]	2.25 [2.09, 2.41]	2.11 [1.91, 2.32]	2.26 [2.00, 2.52]	2.37 [2.02, 2.71]	2.42 [1.96, 2.87]
Age (years)	2.36 [2.33, 2.40]	2.41 [2.36, 2.45]	2.42 [2.37, 2.48]	2.41 [2.34, 2.48]	2.45 [2.36, 2.54]	2.48 [2.37, 2.58]	2.50 [2.37, 2.64]
Body Mass Index	0.49 [0.44, 0.55]	0.52 [0.45, 0.58]	0.51 [0.42, 0.59]	0.54 [0.43, 0.65]	0.56 [0.42, 0.70]	0.51 [0.33, 0.69]	0.47 [0.23, 0.71]
Baseline CDR-SB Score	-0.02 [-0.04, -0.01]	-0.04 [-0.06, -0.02]	-0.06 [-0.08, -0.04]	-0.08 [-0.11, -0.05]	-0.13 [-0.17, -0.09]	-0.15 [-0.20, -0.10]	-0.17 [-0.23, -0.10]
Paid Job vs No Paid Job	-18.33 [-19.09, -17.57]	-19.24 [-20.17, -18.30]	-19.77 [-20.97, -18.56]	-19.01 [-20.54, -17.47]	-20.19 [-22.13, -18.25]	-20.86 [-23.30, -18.41]	-21.07 [-24.18, -17.96]
Sex: Female vs Male	-0.28 [-0.51, -0.05]	-0.40 [-0.68, -0.12]	-0.18 [-0.54, 0.18]	-0.22 [-0.68, 0.25]	-0.34 [-0.94, 0.26]	-0.37 [-1.15, 0.41]	-0.57 [-1.64, 0.49]
Baseline Tau PET	0.43 [-0.57, 1.43]	0.37 [-2.34, 3.09]	-4.06 [-11.47, 3.35]	NA	NA	NA	NA
Lives in Nursing Home vs Home	1.69 [1.43, 1.94]	1.94 [1.63, 2.25]	2.10 [1.69, 2.51]	2.24 [1.69, 2.80]	1.94 [1.18, 2.71]	1.43 [0.34, 2.52]	2.16 [0.42, 3.90]

- The biomarker effects (amyloid and tau PET) jump around from year to year and their CIs are often wide or include zero.
- The other variables shows more consistent effects across years.

Conclusions and modeling implications

- BPRS increases steadily over time; subgroups mainly differ by level (sex, job, living situation, centre), not by shape of the curve.
- Age shows a roughly linear relation with BPRS, while BMI has a curved (non-linear) relation.
- Dropout is higher in patients with worse baseline BPRS.
- For population averages we use a repeated-measures model with an unstructured covariance; for patient-specific effects we use a mixed model with patient random intercept and slope plus a trial random intercept.
- We do not recommend including TAUPET0 or ABPET0 in longitudinal models: TAUPET0 has almost no variation after year 3, and ABPET0 shows an unstable pattern with BPRS over time.

Two-Stage Model Results

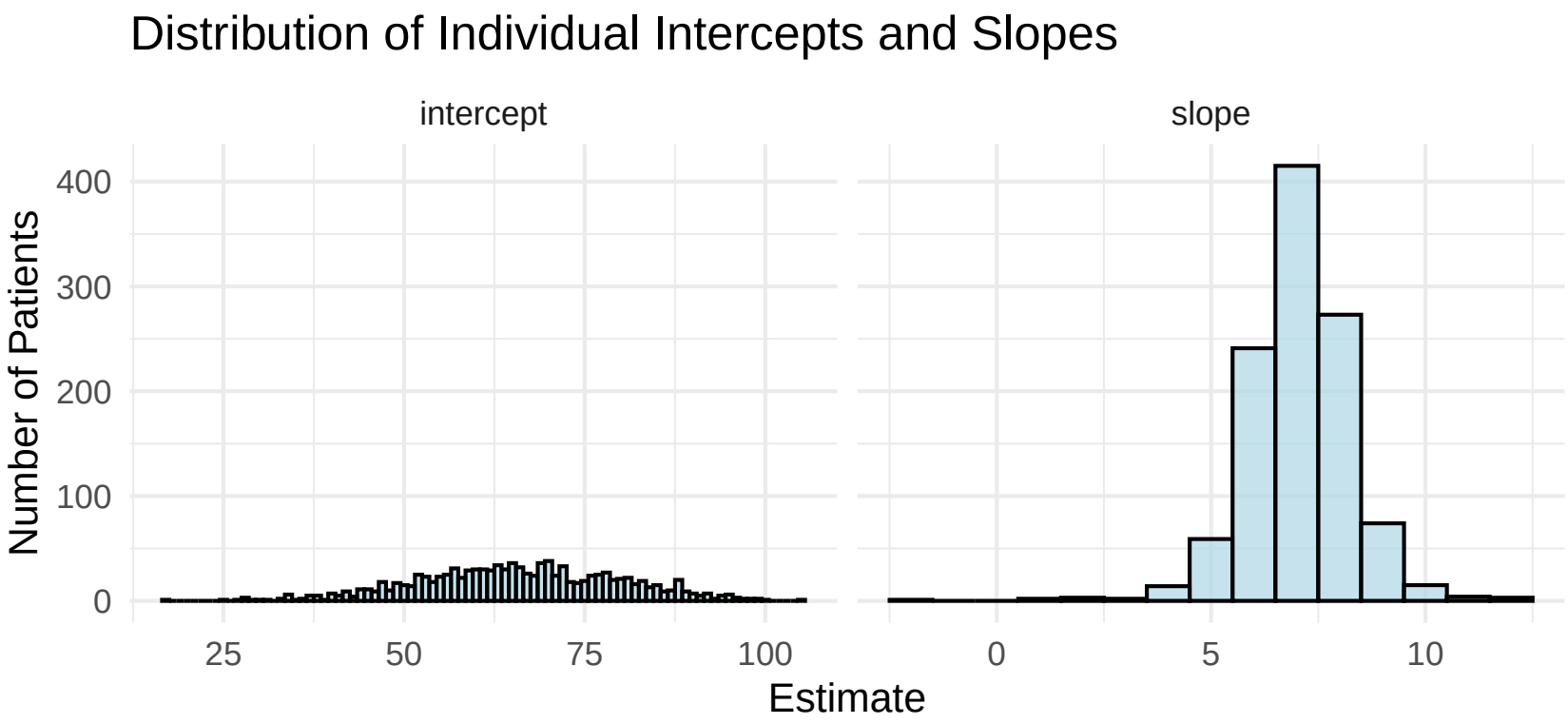

Figure 4: Distribution of Individual Intercepts and Slopes from two stage analysis

- Starting levels vary a lot Intercepts are widely spread (20–100 BPRS), showing big differences in baseline severity between patients.
- Progression rates are similar Slopes are tightly clustered around 6- 7 points/year, so most patients worsen at a similar rate.
- Bias Patients with higher BPRS are more likely to drop out, the slopes mainly reflect those who stay in follow-up.

Table 2: Two-Stage Model Results for Individual Slopes

term	estimate	std.error	statistic	p.value
(Intercept)	6.37 [2.97, 9.77]	1.73	3.68	<0.01
sexFemale	-0.04 [-0.19, 0.11]	0.08	-0.55	0.58
age	0.01 [-0.01, 0.04]	0.01	1.02	0.31
bmi	0.00 [-0.03, 0.04]	0.02	0.09	0.93
jobPaid Job	-0.34 [-0.85, 0.17]	0.26	-1.32	0.19
adl	0.05 [-0.02, 0.11]	0.03	1.37	0.17
wzcNursing Home	0.19 [0.02, 0.36]	0.09	2.23	0.03
cdrsb0	-0.02 [-0.03, -0.01]	0.01	-4.30	<0.01
abpet0	-0.07 [-0.34, 0.20]	0.14	-0.53	0.60
taupet0	-0.13 [-1.61, 1.34]	0.75	-0.17	0.86

- Only two baseline factors show an association with the rate of change patients living in a nursing home worsen slightly faster (about 0.2 points more per year), while those with higher baseline CDR-SB show a slightly slower increase in BPRS (about 0.02 points less per year per CDR-SB point).

Mixed Model with Random Intercept and Slope Results

Table 3: Mixed Model with Random Interceptand Slope Results

effect	term	estimate	std.error	statistic	p.value
fixed	(Intercept)	-60.29 [-68.48, -52.11]	4.17	-14.46	<0.01
fixed	time	7.22 [7.14, 7.30]	0.04	179.32	<0.01
fixed	sexFemale	-0.16 [-0.33, 0.01]	0.09	-1.84	0.07
fixed	age	1.70 [1.62, 1.78]	0.04	42.36	<0.01
fixed	bmi	0.18 [0.12, 0.23]	0.03	6.54	<0.01
fixed	jobPaid Job	-5.14 [-6.71, -3.56]	0.80	-6.39	<0.01
fixed	adl	0.07 [-0.17, 0.30]	0.12	0.55	0.58
fixed	wzcNursing Home	1.92 [1.72, 2.11]	0.10	19.56	<0.01
fixed	cdrsb0	0.00 [-0.01, 0.02]	0.01	0.26	0.79
fixed	time:cdrsb0	-0.02 [-0.03, -0.01]	0.00	-5.25	<0.01

- On average, BPRS increases by about ≈ 7.2 points per year (time effect), confirming steady worsening over follow-up.
- Higher age, higher BMI, living in a nursing home, and not having a paid job are all associated with higher BPRS levels at any given time.
- The interaction time $\times$ cdrsb0 is negative: patients with worse baseline cognition (higher CDR-SB) show a slightly slower yearly increase in BPRS.

Diagnostics Plots

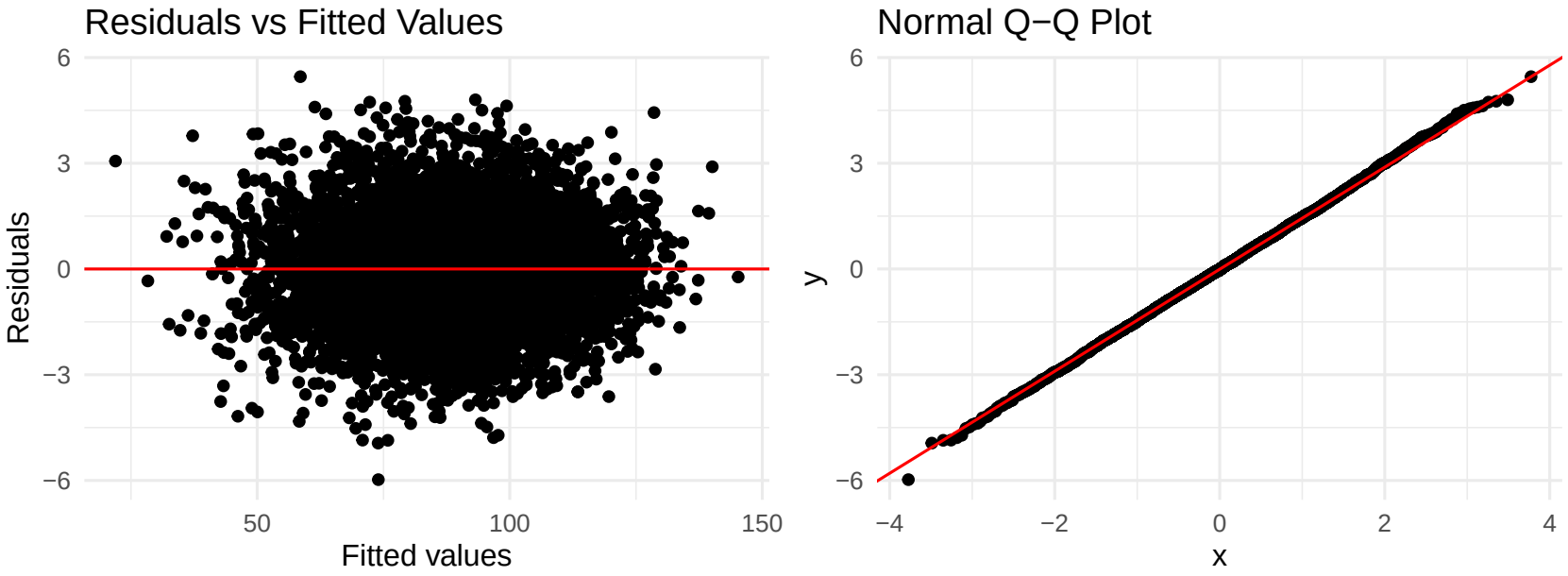

Figure 5: Mixed Model Residual Diagnostics

- Residuals are scattered evenly around zero with no clear pattern, so the linear fit and constant variance look reasonable.
- In the Q–Q plot, points lie close to the straight line, so the normality assumption for residuals is roughly satisfied (only small deviations in the tails).

Comparing with Two Stage Results

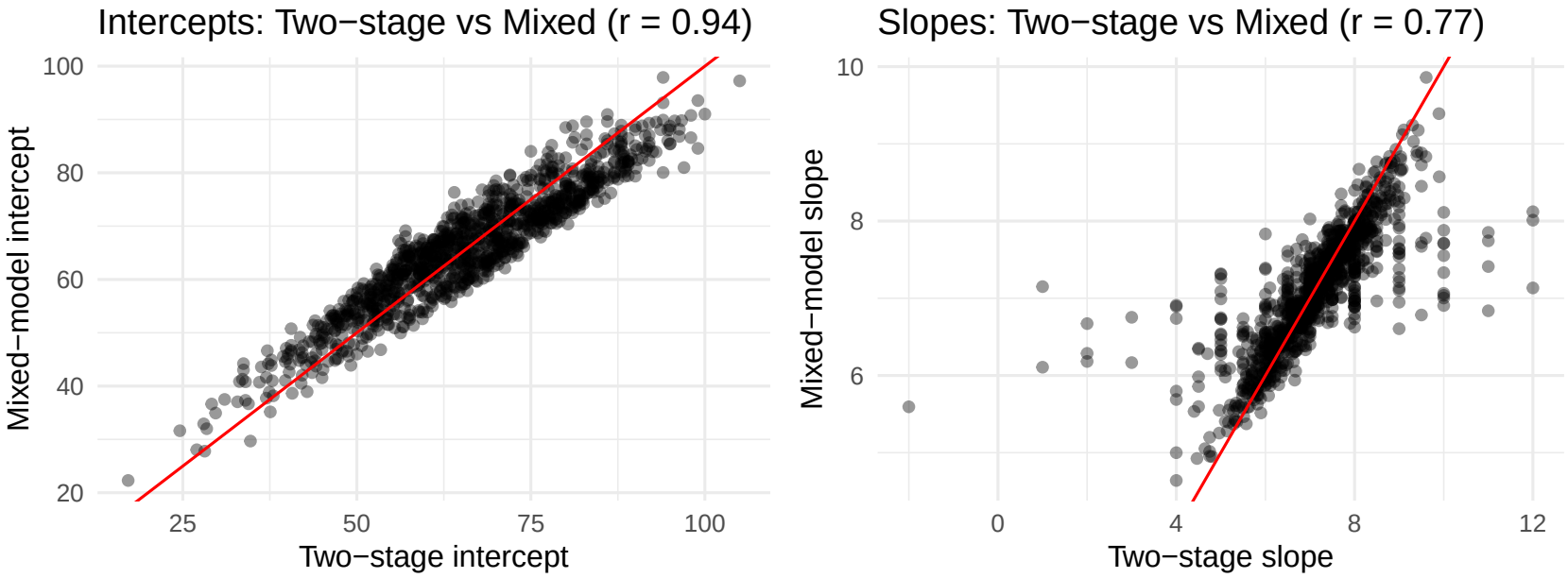

Figure 6: Comparison of Two-Stage and Mixed Model Estimates

- Intercepts from the two-stage and mixed models are highly correlated (r = 0.94), so both approaches give almost the same starting BPRS level.
- Slopes correlate well but less strongly (r = 0.77); the mixed model pulls extreme slopes towards the average.

Recommendations for future similar studies

Based on what we observed in the exploratory plots and models, we suggest the following design and data collection improvements for future work on BPRS in Alzheimer patients.

1. Address informative dropout and missingness: Correlation of missngness and baseline BPRS
2. Standardise clinical site procedures High variability between sites